

De re readings of the Lobi logophoric pronoun: results and diagnostics¹

Shweta Akolkar — UC Berkeley

Sansan Claude Hien — UC Berkeley

Abstract. It is debated whether logophoric pronouns across languages are obligatorily construed de se. Much of the debate has centered on a single language, Ewe (Pearson 2015, Bimpeh 2019, Bimpeh 2023, Bassi et al. 2023, i.a.). This study reports on the finding that the logophoric pronoun in Lobi, another West African language, is compatible with de re readings. We reflect on the methodological recommendations in earlier works for de re testing and suggest that they may introduce biases and unwanted readings. In light of this, we offer new methodological suggestions.

1 Introduction

Languages of the Macro-Sudan belt commonly have logophoric pronouns in addition to ordinary pronouns (Güldemann 2003). Logophoric pronouns occur in propositional attitude environments, such as reports of speech and thought, and obligatorily refer to the attitude holder. A frequently cited example from Ewe is repeated in (1).

(1) *Logophoric and ordinary pronouns in Ewe* (Clements 1975: 142)

- | | |
|---|---|
| a. Kofi be <i>yè</i> -dzo.
Kofi say LOG-leave
'Kofi _i said that he _i left.' | b. Kofi be <i>e</i> -dzo.
Kofi say 3SG-leave
'Kofi _i said that (s)he _j left.' |
|---|---|

The logophoric pronoun *yè* in (1a) refers to the reported speaker, Kofi, while the ordinary third person pronoun *e* in (1b) has disjoint reference.

There is ongoing debate regarding the precise interpretation of logophoric pronouns, particularly in terms of whether they are obligatorily read de se. A de se expression in an attitude report not only refers to the attitude holder but also requires that the attitude holder be aware of their self-reference during the reported attitude event. Many expressions across languages—including logophoric pronouns (e.g. Anand 2006), shifted indexicals (e.g. Schlenker 1999), and long-distance reflexives (e.g. Oshima 2006)—have been claimed to be restricted to de se construal.

The hypothesis that logophoric pronouns are obligatory de se expressions was initially pursued on the basis of a predicted parallel with obligatorily controlled PRO, prior to thorough empirical testing of their interpretations (Schlenker 1999, Heim 2002, von Stechow 2003, i.a.). In the first detailed investigation of the construal of the Ewe logophor *yè*, however, Pearson (2015) found the pronoun to be compatible with de re readings. Her results have since been replicated in a few studies on different varieties of Ewe (O'Neill 2015, Bimpeh 2023, Satik 2021) but contested by

¹We would like to thank Itai Bassi, Abigail Bimpeh, Amy Rose Deal, and audience members at ACAL 55 and TripleA 11 for discussions related to this work. To our consultant who confirmed the Lobi judgments in this paper, but preferred to remain anonymous, *nré hína fwór!*

others (Bimpeh 2019, Bassi et al. 2023). The latter works argue that *yè* is indeed restricted to *de se* readings, with Bassi et al. (2023) further extending this claim to logophoric pronouns in Yoruba and Igbo. These findings are consistent with earlier studies on logophors in Bafut (Kusumoto 1998), Yoruba (Adesola 2005, Anand 2006), and Tangale (Haida 2009).

In this literature, the differences in the findings on Ewe *yè* have largely been attributed to the methodological difficulties of *de re* testing, rather than to dialectal or interspeaker variation. Researchers have highlighted that the scenarios used to test for *de re* readings can be confusing for consultants (Bimpeh 2019), and may inadvertently introduce biases toward one interpretation over the other (Pearson 2022, Bassi et al. 2023). To accurately analyze logophoric pronouns across languages and understand how the dependency between these pronouns and their antecedents is encoded in the grammar, it is crucial to develop a more robust methodology for diagnosing *de re* readings and to apply this methodology to a larger sample of languages with logophoric pronouns.

This study brings data from Lobi (Gur) to bear on the issue of the interpretation of logophoric pronouns. To our knowledge, our paper provides the first detailed description of logophors in the language.² We find that the Lobi logophoric pronoun is compatible with *de re* readings and outline a recommended methodology for testing this property in other languages, with the aim of contributing to the development of best practices in *de re* testing. The paper proceeds as follows. Section 2 provides an overview of the Lobi language and our data sources, as well as a description of the basic properties of the logophoric pronoun. Section 3 discusses standard approaches to *de re* testing, recent proposals for improving this methodology, and the challenges presented by each, ultimately motivating our own modified methodology. Section 4 presents the results of our testing in Lobi, and Section 5 concludes with a discussion of the implications of our results for the interpretations of logophoric pronouns across languages.

2 Preliminaries

2.1 Language background

Lobi belongs to the Gur (or Mabilia) branch of the Atlantic-Congo family. Traditionally, the language has been centered around the town of Bouna in northeastern Côte d’Ivoire, with significant speaker populations in neighboring regions of Burkina Faso and Ghana. In recent years, Lobi communities have spread to other parts of Côte d’Ivoire and Ghana as well. According to Ethnologue (Eberhard et al. 2022), there are approximately 490,000 speakers as of 2009.

2.2 Data & contributions

The data in this paper come from two speakers. The second author of the paper has served as the primary consultant. Judgments with respect to *de se* and *de re* construal were collected in a series of eight elicitation sessions between the first and second author. Scenarios were presented multiple times across across different sessions to ensure the accuracy and consistency of the judgments. Sometimes small adjustments were made to the scenarios or attitude verbs in the target sentences

²A few grammatical descriptions of Lobi (Lamothe 1966, Vaillant 1967, Becuwe 1982) mention a logophoric or reported speech pronoun but do not discuss its properties in detail.

to determine whether these changes influenced the results. A set of finalized scenarios was then formulated into a questionnaire which the second author administered in French to another native Lobi speaker via two phone elicitation sessions.

Both Lobi consultants were born and raised in Agnibilékrou, Côte d’Ivoire. The second author currently resides in Northern California while our remote consultant continues to live in Agnibilékrou.

2.3 Properties of the logophoric pronoun

Lobi has two pronouns that are used for third-person singular reference: *ón* and *á*. This section demonstrates that the former has the distribution of a logophoric pronoun, while the latter is generally restricted to non-logophoric contexts. As shown in Table 1, the form of these pronouns varies depending on their syntactic position. We refer to them by their (nominative) citation forms, given in the top row of the table.

	Logophoric pronoun	Ordinary pronoun
Subject	<i>ón</i>	<i>á</i>
Object (direct, indirect, applied)	<i>n(ĩ)</i>	<i>V, wó</i> ³
Possessor	<i>ón/ónĩ</i>	<i>óó</i>

Table 1: Third person singular pronoun forms in Lobi

Lobi *ón* occurs in the complements of attitude verbs, where it obligatorily refers to the attitude holder (2).⁴

- (2) a. oló só-r [ón gáá-lé]. b. oló jínā-né [irí jí-nĩ-né].
 Olo say-FIN LOG leave-FIN Olo remember-FIN Iri see-LOG-FIN
 ‘Olo_i said that he_{i,*j} left.’ ‘Olo_i remembered that Iri saw him_{i,*j}.’

When *á* occurs under an attitude verb, by contrast, it cannot refer to the attitude holder (3). (This disjointness requirement is weakened in cases of multiple embedding, as discussed in the Appendix.)

³The ordinary object pronoun has varied realizations. In typical root clauses, it occurs as a vowel segment that assimilates to the qualities of the immediately preceding vowel of the verb stem, as seen in (3b). In other environments (not shown in this paper), it forms an independent phonological word and is realized as *wó*.

⁴The following glossing abbreviations are used in Lobi examples throughout the paper: 1, 2, 3 = first, second, and third persons; CC = clause connector; DEF = definite; FIN = finite; FUT = future tense; LOG = logophoric pronoun; NEG = negation; PST = past tense; PL = plural; POSS = possessive; PROG = progressive aspect; REFL = reflexive; SG = singular.

- (3) a. oló só-r [á gáá-lé].
 Olo say-FIN **3SG** leave-FIN
 ‘Olo_i said that he_{*i,j} left.’
 b. oló jínã-né [irí jí-í-ré].
 Olo remember-FIN Iri see-**3SG**-FIN
 ‘Olo_i remembered that Iri saw him_{*i,j}.’

Pronominal *ún* cannot occur in non-attitudinal environments, such as matrix clauses (4), non-attitudinal embedded clauses (5, 6), or non-clausal complements (7). In these environments, *á* must be used instead.

- (4) a. ***ún** gáá-lé.
LOG leave-FIN
 Intended: ‘(S)he left.’
 b. **á** gáá-lé.
3SG leave-FIN
 ‘(S)he left.’
- (5) a. *bisààn-ná gáá-lé [sɪ **ún** jí **ún** ní].
 child-DEF leave-FIN before **LOG** see **LOG**.POSS mother
 Intended: ‘The child_i left before he_i saw his_i mother.’
 b. bisààn-ná gáá-lé [sɪ **á** jí **ɔ̀** ní].
 child-DEF leave-FIN before **3SG** see **3SG**.POSS mother
 ‘The child_i left before he_i saw his_i mother.’
- (6) a. *irí cóú-rɛ [dɪ oló jí-**ní**].
 Iri make-FIN CC Olo see-**LOG**
 Intended: ‘Iri_i made Olo see her_i.’
 b. irí cóú-rɛ [dɪ oló jí-**í**].
 Iri make-FIN CC Olo see-**ORD**
 ‘Iri_i made Olo see her_i.’
- (7) a. *oló nã kpíérá-r **ún** ní.
 Olo PROG think.about-FIN **LOG**.POSS mother
 Intended: ‘Olo_i is thinking about his_i mother.’
 b. oló nã kpíérá-r **ɔ̀** ní.
 Olo PROG think.about-FIN **3SG**.POSS mother
 ‘Olo_i is thinking about his_i mother.’

Additionally, Lobi *ún* may not be coreferential with any long-distance nominal expression that does not denote an attitude holder. For instance, it cannot be anteceded by the goal argument of the matrix verb *míé* ‘tell’ in (8).

- (8) irí míé-rɛ oló [**ún** gbáá-nē].
 Iri tell-FIN Olo **LOG** be.tall-FIN
 ‘Iri_i told Olo_j that she_{i,*j,*k} is tall.’

Given that *ún* is restricted to attitude environments and requires an attitude holder DP as its antecedent, we classify it as a logophoric pronoun. Meanwhile, the wider distribution of *á* supports its characterization as an ordinary third person pronoun.

The attitude verbs that we have found to license the logophoric pronoun are listed in Table 2.⁵

Lobi verb	English gloss
<i>só</i>	‘say’
<i>ji</i>	‘say’
<i>míé</i>	‘tell’
<i>blé</i>	‘ask’
<i>káɲmã</i>	‘think’
<i>ji</i>	‘know’
<i>jinã</i>	‘remember’
<i>nîê</i>	‘hear’
<i>na</i>	‘want’
<i>tú</i>	‘agree’

Table 2: Logophor-licensing predicates

When embedded under multiple attitude predicates, the Lobi logophoric pronoun behaves similarly to logophoric pronouns in Ewe (Clements 1975, Pearson 2015) and Yoruba (Anand 2006). First, it may take a long-distance antecedent (9):

- (9) ɓɛɓɛ ɟínã-ně [oló jí [úñ nà-í irí]].
 Bebe remember-FIN Olo say LOG like-FIN Iri
 ‘Bebe_i remembered that Olo_j said that he_{i,j} likes Iri.’

Second, clausemate logophoric pronouns need not have the same antecedent. When two instances of *úñ* occur in the same multiply embedded attitude report, each can be coreferential with a different attitude holder DP (10).

- (10) ɓɛɓɛ ɟínã-ně [irí só-r [úñ gbá-r úñ mʉ́sú]].
 Bebe remember-FIN Iri say-FIN LOG take-FIN LOG.POSS money
 ✓ ‘Bebe_i remembered that Iri_j said that he_i took his_i money.’
 ✓ ‘Bebe_i remembered that Iri_j said that she_j took her_j money.’
 ✓ ‘Bebe_i remembered that Iri_j said that he_i took her_j money.’
 ✓ ‘Bebe_i remembered that Iri_j said that she_j took his_i money.’

A final observation is that the distinction between logophoric and ordinary pronouns is made only in the third-person singular. The pronoun *úñ* can only have a third-person singular antecedent (11), and pronouns of other persons and numbers do not have distinct logophoric counterparts (12).

⁵Of the attitude verbs we considered, only *díí* ‘dream’ was incompatible with *úñ* in its complement. We also considered predicates describing emotional states, with translations like ‘be happy/angry/surprised’, but found that they cannot take clausal complements.

- (11) *mɪ/fɪ/wʊ só-r [ʊ́h gáá-lé].
1SG/2SG/3PL say-FIN **LOG** leave-FIN

Intended: ‘{I, you, they}_i said that {I, you, they}_i left.’

- (12) a. mɪ só-r [mɪ gáá-lé]. b. fɪ só-r [fɪ gáá-lé].
1SG say-FIN **1SG** leave-FIN 2SG say-FIN **2SG** leave-FIN
‘I said that I left.’ ‘You_i said that you_i left.’
- c. wʊ só-r [wʊ gáá-lé].
3PL say-FIN **3PL** leave-FIN
‘They_i said that they_i left.’

In sum, the behavior of Lobi ʊ́h is fairly typical for a logophoric pronoun. We now turn to a discussion of its interpretation.

3 Diagnosing de re readings

3.1 Mistaken identity and its challenges

It is undisputed that logophoric pronouns are compatible with de se attitudes, as illustrated for Lobi in (13), a sentence which clearly reports a self-ascribed belief.

- (13) Scenario: *Thangba has the thought: “Ini likes me.”*

thángbá káɲmã [iní ná-nĩ-ně́].
Thangba think Ini like-**LOG**-FIN

‘Thangba_i thinks that Ini likes him_i.’

Whether logophoric pronouns also have de re readings can be determined on the basis of mistaken or unknown identity scenarios, where an attitude holder does not recognize that their belief is about themselves. Let’s consider such a scenario from Schlenker (1999):

- (14) Scenario: *John, who is a candidate in the election, is so drunk that he doesn’t remember who he is. He watches TV and sees a candidate he finds terrific, thinking that this guy must be a hero. This candidate happens to be John himself, though he doesn’t realize it.* (Schlenker 1999, p. 97)

In this scenario, John does not self-ascribe the property of being a hero: therefore his belief is non-de se. If in a given language, the sentence corresponding to “John thinks LOG is a hero” can be used to truthfully report John’s belief in (14), then the logophoric pronoun in that language must be compatible with de re readings. On the other hand, if the sentence is false in this scenario, then the logophoric pronoun must be restricted to de se readings.

Although such scenarios effectively illustrate the logical distinction between de se and non-de se interpretations of pronouns, they have proven challenging to employ in elicitation settings. I

highlight two key difficulties in testing for de re readings using mistaken identity scenarios. First, if a logophoric pronoun is compatible with de re readings, that means it is *ambiguous* between de se and de re interpretations.⁶ In a mistaken identity scenario, a target sentence containing an ambiguous logophoric pronoun is both true *and* false—true on the de re reading but false on the de se reading. Depending on the scenario, the de se reading may be much more salient, leading consultants to incorrectly judge the sentence as simply false. We can appreciate this by considering the English sentence “John_i thinks he_i is a hero” in the context of the scenario in (14). Although *he* undoubtedly has de re readings, the sentence seems true only in a technical sense; it feels like an unnatural and even misleading way to report John’s belief. In elicitation contexts, one could imagine that some English speakers might focus on the more natural de se reading and, as a result, judge the sentence to be false. A second challenge, noted by Bassi et al. (2023), is that if a mistaken identity scenario is sufficiently vague, it may give consultants leeway to mentally enrich the scenario and make it compatible with a de se reading of the target sentence. For example, the scenario in (14) provides no information about John’s de se beliefs regarding his character. A consultant could mentally fill in this gap and imagine John to self-ascribe being a hero, then judge the target sentence to be true on this unwanted reading.

As these two challenges demonstrate, mistaken identity scenarios have the potential to introduce biases and unwanted readings. For reliable de re testing, it is necessary to minimize these effects.

3.2 Methodology of Bassi et al. (2023)

Bassi et al. (2023), building on previous work by the same authors (Bimpeh 2019, Bimpeh 2023, Bimpeh et al. 2024), propose a modified methodology for de re testing to address the difficulties associated with mistaken identity scenarios. Their recommendations are summarized in (15).

(15) *Summary of the methodological recommendations from Bassi et al. (2023)*

1. Two minimally different target sentences should be tested for compatibility with each scenario: one containing a logophoric pronoun and another containing an ordinary pronoun.
2. Scenarios should ensure that the target sentences are false on de se readings.

The first recommendation establishes a baseline for consultants, allowing them to assess if one type of pronoun is preferred to the other; it also allows the researcher to confirm the consultants’ understanding of the scenario. The second recommendation ensures that consultants do not embellish the context to make it compatible with a de se reading. Following this methodology, Bassi et al. conclude that logophoric pronouns in Ewe, Yoruba, and Igbo are obligatorily read de se, and that de re coreference is achieved using ordinary pronouns. The authors suggest that previous studies may have yielded unreliable results due to the absence of these controls.

While we recognize the benefits of Bassi et al.’s approach, we identify two potential issues in the implementation of their recommendations. These issues are discussed in the following

⁶This could be either because it allows both a dedicated de se representation and a more general de re representation (see e.g. Anand 2006), or because it allows just a de re representation but with a variable over acquaintance relations that can be specified in either a de se or non-de se way (see e.g. Maier 2009).

subsections. Section 3.3 then presents the approach we adopted for de re testing in Lobi, which builds on Bassi et al.'s suggestions while addressing the identified complications.

3.2.1 Issue 1: Exacerbating the de se bias

We anticipated that Bassi et al.'s second recommendation of ensuring that the target sentence is false on a de se reading might inadvertently amplify the issue of biasing speakers toward de se interpretations. In particular, if this recommendation is implemented by specifying a de se belief of the attitude holder that contradicts the de se reading of the target sentence, it could serve as a further indicator of the relevance of the de se reading. Consider the scenario in (16), a variant of (14) modified to adhere to Bassi et al.'s second recommendation.

- (16) Modified version of (14): *John, who is a candidate in the election, is so drunk that he doesn't remember who he is. He watches TV and sees a candidate he finds terrific, thinking that this guy must be a hero. This candidate happens to be John himself, though he doesn't realize it. He even says "I wish I was like that guy, but I'm such a failure!"*

In this modified scenario, John still holds a de re belief about himself being a hero. However, the added statement expressing John's de se belief that he is *not* a hero makes his de se perspective seem relevant in evaluating the target sentence. As a result, it feels quite difficult to judge even the English sentence "John_i thinks he_j is a hero" as true in this scenario, despite the fact *he* clearly has de re readings.

3.2.2 Issue 2: Inviting disjoint readings of the ordinary pronoun

A second potential issue came to our attention during pilot testing. When the second author was presented with typical mistaken identity scenarios and asked to judge the truth of target sentences containing ordinary pronouns, following Bassi et al.'s first recommendation, he occasionally accepted the sentences on unwanted readings. An example is provided in (17).

- (17) Scenario: *Sie is so drunk that he has forgotten he played in a boxing match earlier today. He watches a boxing match on TV and thinks that the person boxing against Olo is terrific and sure to win. Unbeknownst to Sie, the boxer he is watching on TV is Sie himself.*

✓ sèé káɣmã [á ná bé-r oló]

Sie think 3SG FUT beat-FIN Olo

'Sie thinks that he will beat Olo.'

From follow-up discussions, we learned that these sentences were accepted for one of two reasons: first, they could be used felicitously to report the attitude holder's belief from the attitude holder's own perspective, according to which the belief is not about themselves; or second, they could be uttered by someone who is also unaware that the belief is about the attitude holder. In both cases, the speaker of the utterance is treating the attitude holder and pronominal referent as disjoint. As evidence that the ordinary pronoun was not read as coreferential with the attitude holder DP, whenever the second author accepted target sentences containing the ordinary third person pronoun, he also accepted sentences replacing the pronoun with a description like *ke tibil* 'this person' (e.g. 'Sie thinks that *this person* will beat Olo.').

3.3 Our recommended methodology

Up to this point, we have identified several challenges associated with testing for de re readings. We now suggest that one issue potentially underpins them all: the absence of a defined conversational context for the target sentence. In previous studies testing for de re readings of pronouns, consultants were typically asked to judge the truth of target sentences in mistaken identity scenarios. A common way for consultants to approach this type of task is to consider whether they can imagine a conversational context in which the sentence could be uttered felicitously. Because no specific conversational context is provided, the task relies on consultants being able to mentally construct a context and suitable question under discussion to license the de re reading. We believe that a more well-defined task is needed to obtain reliable results.

In our investigation of the Lobi logophoric pronoun, we addressed this issue by presenting consultants with **two-part scenarios** that describe both an attitude context and an utterance context. The first part introduces the mistaken identity, while the second establishes an utterance context in which the attitude holder’s awareness of their self-reference is not at-issue. This latter component is intended to provide a natural context for reporting a de re belief, where the speaker has a communicative goal other than to convey the attitude holder’s first-personal perspective. That is, we provide a conversational context in which the de re reading addresses the question under discussion (in line with suggestions in Pearson 2022).

We also followed both recommendations from Bassi et al. (2023) in our testing: the attitude contexts within our scenarios ensure that the target sentences are false on de se readings, and we asked consultants to provide judgments on sentences containing both logophoric and ordinary pronouns. To prevent the unwanted disjoint reading of the ordinary pronoun discussed in §3.2.2, we made sure to clarify in our utterance contexts that the speaker is aware of the attitude holder’s mistaken identity. Our recommendations are summarized in (18).

(18) *Structure of our recommended two-part mistaken identity scenarios:*

1. Attitude context: Motivates the mistaken identity and ensures that the target sentence is false on a de se reading
2. Utterance context: Establishes the speaker’s awareness that the attitude report is about the attitude holder and ensures that the truth of the target sentence de se is not at-issue at the time of its utterance

4 Testing for de re readings of the Lobi logophoric pronoun

4.1 Scenarios and judgments

Three of the six scenarios included in our questionnaire are presented in (19)-(21), along with the accompanying target sentences. Consultants were asked whether each target sentence could be uttered truthfully and felicitously by the utterance speaker described in the scenario. Both consultants provided identical judgments, so individual responses are not listed separately.

- (19) Scenario 1: *Oho made corn fufu for the first time and brought it to a church event. She didn't have time to taste it beforehand. At the church, she saw that two other people also brought corn fufu. Her friend Naaba convinced her to play a game where she closed her eyes and tried one bite of each person's corn fufu, and then decided which was the best. Oho thought it certainly wouldn't be her own, since she had never made corn fufu before. She tried each dish with her eyes closed and pointed to the one she made saying "this one is the tastiest." She had no idea that she had picked her own dish.*

Right then, Iri, who was watching them play the game from afar, approached them and asked whose corn fufu Oho liked best. Naaba answered:

✓ ɔʔó só-r [úñ ɔʔó ʃur dàá-rɔ bobòí].
Oho say-FIN LOG.POSS corn fufu be.tasty-FIN most

ɔʔó só-r [ʒò ɔʔó ʃur dàá-rɔ bobòí].
Oho say-FIN 3SG.POSS corn fufu be.tasty-FIN most

'Oho_i said that her_i corn fufu is the tastiest.'

- (20) Scenario 2: *Iri loves singing and thinks she sings very well, but in reality, she sounds terrible. Her brother really wants her to hear how bad she is, so he secretly records her singing in the kitchen. He then sends her the recording with the message "She sounds great, doesn't she?" Iri listens to the recording and responds "That girl sounds terrible, she doesn't sing well at all." She doesn't realize it is her own voice in the recording.⁷*

Her brother later tells their mother: "Finally, I got her to say it! ..."

✓ irí jí [úñ=à ʃi nékhi=ã].
Iri say LOG=NEG know singing=NEG

irí jí [á=à ʃi nékhi=ã].
Iri say 3SG=NEG know singing=NEG

'Iri_i said that she_i doesn't sing well.'

- (21) Scenario 3: *Olo reads a poem he wrote 20 years ago and thinks it's wonderful. He does not remember that he wrote it. He says to his wife, "I wonder who wrote this. He is certainly intelligent." He does not imagine that he could have written it, because he does not think of himself as intelligent.*

Olo's wife knows that Olo was commenting on his own poem and is happy that he recognized his talent. She and her friend often talk about Olo's low self-confidence, so later she tells her friend the somewhat surprising news:

⁷Adapted from Bimpeh (2019)

✓ oló jí [úñ kpéé-rɛ] (sɪ á=à ʝ [úñ théthé úñã
 Olo say LOG be.intelligent-FIN but 3SG=NEG know LOG REFL LOG.PROG
 dǎcínǎ]=a).
 compliment=NEG

oló jí [á kpéé-rɛ].
 Olo say 3SG be.intelligent-FIN

‘Olo_i said that he_i is intelligent (but he didn’t know that it’s himself he was compliment-
 ing).’

4.2 Results

Across all of the scenarios, the Lobi consultants found the target sentences containing the logophoric pronoun to be felicitous. Meanwhile, they found sentences with the ordinary pronoun to be infelicitous; even in mistaken identity scenarios, they reiterated that the ordinary pronoun *á* is strictly “for someone else,” unable to refer to the attitude holder. We conclude from these diagnostics that the Lobi logophoric pronoun is compatible with de re readings.

We received positive signals regarding the accuracy of our results. Both consultants were confident and consistent in their judgments. In addition, their comments on Scenarios 1 and 2 demonstrated their comprehension of the task:

- *"In that moment, Oho will learn that she picked her own fufu."* (Scenario 1)
- *"You are talking about Iri, even though she doesn't know it's her."* (Scenario 2)
- *"If Iri hears that, she might say 'I didn't say that' and her brother would have to explain."* (Scenario 2)

For Scenario 3, the target sentence containing the logophor included a second conjunct (“but he didn’t know that it’s himself he was complimenting”) which contradicts the de se reading of the first conjunct. The felicity of this sentence confirms that the logophoric pronoun is interpreted de re.

5 Implications and concluding remarks

Our findings support the hypothesis that logophoric pronouns are not universally restricted to de se interpretations across languages and speakers; for the Lobi consultants in this study, de re readings were readily available. Despite incorporating the methodological recommendations from Bassi et al. (2023) in our testing, our results aligned with Pearson’s rather than theirs. Table 3 summarizes the different findings across these studies.

	Pearson (2015)	Bassi et al. (2023)	This paper
Logophoric pronoun	ambiguous	unambiguously de se	ambiguous
Ordinary pronoun	-	coreference allowed if de re	coreference disallowed

Table 3: De re testing results across studies

We consider three possible explanations for the differences in these results: crosslinguistic variation, interspeaker variation, and methodological inconsistencies.

Let’s first consider the possibility that there are two types of languages with logophoric pronouns: one where the pronoun is ambiguous between de se and de re readings (Type A) and one where it is unambiguously read de se (Type B):

Language Type A			Language Type B		
E.g. Lobi			E.g. Ewe (per Bassi et al. 2023)		
	Coreferent	Non-coreferent		Coreferent	Non-coreferent
De se	LOG	-	De se	LOG	-
De re	LOG	ORD	De re	ORD	ORD

Tables 4 & 5: Two hypothetical types of languages with logophoric pronouns

In both types of languages, the logophoric pronoun would be used in de se attitudes, and the ordinary pronoun would be used for reference to a non-attitude holder. The language types would differ, however, in terms of which pronoun is used to express de re coreference with an attitude holder DP (the boxed cell).

This hypothetical division between languages may arise from different outcomes of competition between logophoric and ordinary pronouns. Let’s assume, following a standard analysis, that logophoric pronouns are bound by individual abstractors positioned at the left edge of attitude clauses (Heim 2002, von Stechow 2003, Pearson 2015), and are therefore interpreted de se by default; in order to be read de re, they require additional structure at LF (Pearson 2015). Meanwhile, ordinary pronouns are individual variables with no binding requirements. Let’s further suppose that the competitive principles of specificity and economy are both relevant to the distributions of these pronouns. On the basis of the semantics we just sketched, it seems plausible that logophoric pronouns would outcompete ordinary pronouns for de se construal because they are more specified, applying in narrower contexts. For de re coreference, however, the competition may be less straightforward: while the logophoric pronoun has a narrower domain, it also requires additional structure to be interpreted de re, making it the less economical choice. Different languages may resolve this competition differently, resulting in crosslinguistic variation.

Pearson (2015; 2022) has challenged the idea that the availability of de re readings for logophors could be a point of crosslinguistic variation, reasoning that mistaken identity scenarios are too rare in ordinary life for a child to receive sufficient input to converge on one grammar over the other. However, as Pearson acknowledges, there may be other properties of logophoric pronouns that act as cues for the availability of de re readings; therefore, crosslinguistic variation cannot be ruled out entirely. Additionally, we believe that Pearson’s reasoning does not rule out the

possibility of interspeaker variation: in the absence of sufficient input, perhaps individual speakers simply settle on one grammar or the other.⁸ Thus, rather than a language-level split, the Type A/B distinction may apply to grammars of individuals. That said, the fact that researchers have generally reported consistent findings across speakers of a language—even in studies with larger samples (e.g., Bimpeh 2023)—suggests that interspeaker variation alone does not account for the discrepancies between studies.

A final possible explanation for the discrepancies in the literature, highlighted throughout this paper, is the lack of a standardized and reliable methodology for diagnosing *de re* readings. Without a methodology that allows for the replication of results across studies, we cannot fully evaluate the extent of crosslinguistic and interspeaker variation in logophoric pronouns and similar expressions. We hope that the approach presented here will prove useful to future researchers conducting *de re* testing and contribute to the development of a standardized methodology.

References

- Adesola, Oluseye. 2005. Pronouns and null operators: A-bar dependencies and relations in Yoruba. Doctoral Dissertation, Rutgers University.
- Anand, Pranav. 2006. *De de se*. Doctoral Dissertation, Massachusetts Institute of Technology.
- Bassi, Itai, Imke Driemel, Abigail Anne Bimpeh, and Silvia Silleresi. 2023. Decomposing logophoric pronouns: a presuppositional account of logophoric dependencies. Manuscript, November 2023.
- Becuwe, Jacques. 1982. *Eléments de phonologie et de grammaire du Lobiri (Parler de Bouna, Cote-d’Ivoire)*. Doctoral Dissertation, Université de la Sorbonne Nouvelle (Paris 3).
- Bimpeh, A, Imke Driemel, Itai Bassi, and Silvia Silleresi. 2024. Obligatory *de se* logophors in Ewe, Yoruba, and Igbo: Variation and competition. In *Proceedings of WCCFL 40*.
- Bimpeh, Abigail Anne. 2019. Default *de se*: The interpretation of the Ewe logophor. In *Proceedings of TripleA 5*.
- Bimpeh, Abigail Anne. 2023. Logophoricity in Ewe: An empirical-semantic assessment of *yè*. Doctoral Dissertation, Doctoral dissertation, Johann Wolfgang Goethe-Universität zu Frankfurt am Main.
- Clements, George N. 1975. *The logophoric pronoun in Ewe: Its role in discourse*.
- Eberhard, D., M. G. F. Simons, and C. D. Fennig. 2022. Lobi – Ethnologue: Languages of the world. [https://www-ethnologue-com.libproxy.berkeley.edu/language/lob](https://www.ethnologue-com.libproxy.berkeley.edu/language/lob).
- Güldemann, Tom. 2003. Logophoricity in Africa: An attempt to explain and evaluate the significance of its modern distribution. *STUF-Language Typology and Universals* 56:366–387.
- Haida, Andreas. 2009. Logophoricity in Tangale. In *40th Annual Meeting of the North East Linguistic Society (NELS 40)*, Cambridge, MA, Nov, 13–15.
- Heim, Irene. 2002. Features of pronouns in semantics and morphology. Handout of a talk given at USC.

⁸Thanks to Amy Rose Deal for suggesting this possibility.

- Kusumoto, Kiyomi. 1998. Tenses as logophoric pronouns. In *Handout of talk given at the MIT/UConn/UMass Semantics Workshop*.
- Lamothe, Charles PB. 1966. *Esquisse du système grammatical Lobi*. CNRS ua.
- Maier, Emar. 2009. Presupposing acquaintance: A unified semantics for de dicto, de re and de se belief reports. *Linguistics and Philosophy* 32:429–474.
- Oshima, David Yoshikazu. 2006. Perspectives in reported discourse. Doctoral Dissertation, Stanford University Stanford.
- O'Neill, Teresa. 2015. The distribution of the Danyi Ewe logophor yi. In *Talk given at the LSA Annual Meeting*.
- Pearson, Hazel. 2015. The interpretation of the logophoric pronoun in Ewe. *Natural Language Semantics* 23:77–118.
- Pearson, Hazel. 2022. On testing for de se and de re construals across languages. *Semantic Field Methods* 4:1–22.
- Satik, Deniz. 2021. Control is not movement: Evidence from overt PRO in Ewe. *Unpublished manuscript*.
- Schlenker, Philippe. 1999. Propositional attitudes and indexicality: A cross categorial approach. Doctoral Dissertation, Massachusetts Institute of Technology.
- von Stechow, Arnim. 2003. Feature deletion under semantic binding: Tense, person, and mood under verbal quantifiers. In *Proceedings of the Thirty-Third Annual Meeting of the Northeast Linguistics Society*, volume 33, 379–404.
- Vaillant, Par M. 1967. *Esquisse grammaticale du Lobiri*. Doctoral Dissertation, Université de Dakar.

Appendix: More on long-distance logophors

The disjointness requirement on the ordinary pronoun is weakened in contexts of long-distance antecedence: either a logophoric *or* ordinary pronoun may be used for coreference with a long-distance attitude holder DP (22).

- (22) khér-rá jínã-ně [mí só-r [má hé-nĩ-ně / hí-ĩ-ré]].
 woman-DEF remember-FIN 1SG say-FIN 1SG.FUT marry-LOG-FIN / marry-3SG-FIN
 ‘The woman_i remembered that I said that I would marry her_i.’

One may wonder if, given this optionality, the choice of pronoun in long-distance contexts is governed by the de se/de re distinction. We tested this hypothesis with the mistaken identity scenario in (23).

- (23) *Scenario: Olo was just selected for the Emerging Writer award and his picture was published in a magazine talking about the award. He has no idea that he won the award or that his picture appeared in the magazine. He overhears Sie, his senior colleague who does not know or recognize Olo, say that the Emerging Writer award winner is ugly—a comment he has made based on the magazine picture. Olo tells his coworker Ini: “Sie said that the Emerging Writer award winner is ugly.”*

Ini knows that the award winner is Olo. When she's asked by another colleague "What does Sie think about Olo as the award winner?" she answers:

✓ oló jí [sèé só-r [úñ dèé-ré]].
Olo say Sie say-FIN LOG be.ugly-FIN

✓ oló jí [sèé só-r [á dèé-ré]].
Olo say Sie say-FIN 3SG be.ugly-FIN

‘According to Olo_i, Sie said that he_i is ugly.’ (Lit: ‘Olo_i said that Sie said that he_i is ugly.’)

Even in long-distance contexts, the logophoric pronoun is not reserved exclusively for de se construal. What precisely governs the choice between a logophoric and an ordinary pronoun in these environments remains to be determined.